The procedure for finding the terms of a sequence in a recursive manner is called **recurrence relation**. We study the theory of linear recurrence relations and their solutions. Finally, we introduce generating functions for solving recurrence relations.

Definition

A recurrence relation is an equation that recursively defines a sequence where the next term is a function of the previous terms (Expressing F_n Fn as some combination of F_i Fi with $i < n$ i<n).

Example – Fibonacci series – $F_n = F_{n-1} + F_{n-2}$ Fn=Fn−1+Fn−2, Tower of Hanoi – $F_n = 2F_{n-1} + 1$ Fn=2Fn−1+1

Linear Recurrence Relations

A linear recurrence equation of degree k or order k is a recurrence equation which is in the format

$x_n = A_1 x_{n-1} + A_2 x_{n-1} + A_3 x_{n-1} + \ldots A_k x_{n-k}$ xn=A1xn−1+A2xn−1+A3xn−1+…Akxn−k(A_n An is a constant and $A_k \neq 0$ Ak≠0) on a sequence of numbers as a first-degree polynomial.

These are some examples of linear recurrence equations –

Recurrence relations	Initial values	Solutions
$F_n = F_{n-1} + F_{n-2}$	$a_1 = a_2 = 1$	Fibonacci number
$F_n = F_{n-1} + F_{n-2}$	$a_1 = 1, a_2 = 3$	Lucas Number
$F_n = F_{n-2} + F_{n-3}$	$a_1 = a_2 = a_3 = 1$	Padovan sequence
$F_n = 2F_{n-1} + F_{n-2}$	$a_1 = 0, a_2 = 1$	Pell number

How to solve linear recurrence relation

Suppose, a two ordered linear recurrence relation is – $F_n = AF_{n-1} + BF_{n-2}$ Fn=AFn−1+BFn−2 where A and B are real numbers.

The characteristic equation for the above recurrence relation is –

$$x^2 - Ax - B = 0$$
x2−Ax−B=0

Three cases may occur while finding the roots –

Case 1 – If this equation factors as $(x - x_1)(x - x_1) = 0$ (x−x1)(x−x1)=0 and it produces two distinct real roots x_1x1 and x_2x2, then $F_n = ax_1^n + bx_2^n$ Fn=ax1n+bx2nis the solution. [Here, a and b are constants]

Case 2 – If this equation factors as $(x - x_1)^2 = 0$ (x−x1)2=0 and it produces single real root x_1x1, then $F_n = ax_1^n + bnx_1^n$ Fn=ax1n+bnx1n is the solution.

Case 3 – If the equation produces two distinct complex roots, x_1x1and x_2x2 in polar form $x_1 = r\angle\theta$ x1=r∠θ and $x_2 = r\angle(-\theta)$ x2=r∠(−θ), then $F_n = r^n(a\cos(n\theta) + b\sin(n\theta))$ Fn=rn(acos(nθ)+bsin(nθ)) is the solution.

Problem 1

Solve the recurrence relation $F_n = 5F_{n-1} - 6F_{n-2}$ Fn=5Fn−1−6Fn−2 where $F_0 = 1$ F0=1and $F_1 = 4$ F1=4

Solution

The characteristic equation of the recurrence relation is –

$$x^2 - 5x + 6 = 0,$$
x2−5x+6=0,

So, $(x - 3)(x - 2) = 0$ (x−3)(x−2)=0

Hence, the roots are –

$x_1 = 3$ x1=3 and $x_2 = 2$ x2=2

The roots are real and distinct. So, this is in the form of case 1

Hence, the solution is –

$$F_n = ax_1^n + bx_2^n$$
Fn=ax1n+bx2n

Here, $F_n = a3^n + b2^n$ $(\text{As } x_1 = 3 \text{ and } x_2 = 2)$ Fn=a3n+b2n (As x1=3 and x2=2)

Therefore,

$1 = F_0 = a3^0 + b2^0 = a + b$ 1=F0=a30+b20=a+b

$4 = F_1 = a3^1 + b2^1 = 3a + 2b$ 4=F1=a31+b21=3a+2b

Solving these two equations, we get $a = 2$ a=2 and $b = -1$ b=−1

Hence, the final solution is –

$$F_n = 2.3^n + (-1).2^n = 2.3^n - 2^n$$
Fn=2.3n+(−1).2n=2.3n−2n

Problem 2

Solve the recurrence relation – $F_n = 10F_{n-1} - 25F_{n-2}$ Fn=10Fn−1−25Fn−2 where $F_0 = 3$ F0=3 and $F_1 = 17$ F1=17

Solution

The characteristic equation of the recurrence relation is –

$$x^2 - 10x - 25 = 0$$
x2−10x−25=0

So $(x-5)^2 = 0$ (x−5)2=0

Hence, there is single real root $x_1 = 5$ x1=5

As there is single real valued root, this is in the form of case 2

Hence, the solution is –

$F_n = ax_1^n + bnx_1^n$ Fn=ax1n+bnx1n

$3 = F_0 = a.5^0 + b.0.5^0 = a$ 3=F0=a.50+b.0.50=a

$17 = F_1 = a.5^1 + b.1.5^1 = 5a + 5b$ 17=F1=a.51+b.1.51=5a+5b

Solving these two equations, we get $a = 3$ a=3 and $b = 2/5$ b=2/5

Hence, the final solution is – $F_n = 3.5^n + (2/5).n.2^n$